MANDI PRICE WATCH

Detailed Business & Analytics Report

Indian agricultural market intelligence: price levels, volatility and seasonality across 175 crops, 2002–2026

Source data: Agmarknet (Directorate of Marketing & Inspection) via data.gov.in, compiled by Today Market Rates (file built 3 Nov 2026). Each row is a crop-month: the median across markets of each market's average modal price, in ₹ per quintal. All figures in this report were recomputed from the CSV and cross-checked against the DAX measures in the Power BI model.

Executive summary

The dashboard answers four questions for a procurement or market-analysis team: which crops are expensive, which are risky, when prices move, and what to do about it. The headline findings are:

- **Volatility is concentrated in cheap perishables.** Carrot (82.1%), Kinnow (81.4%), Radish (77.9%), Drumstick (74.7%) and Onion (73.7%) top the volatility ranking among crops with at least 12 months of data. The median volatility of low-price-tier crops is 47.8%, against 27.1% for the high-price tier.
- **Higher-priced crops tend to be steadier.** Across 150 crops the rank correlation between median price and volatility is −0.44. This is an association; the data does not show why.
- **Seasonality is strong and has a clear shape.** Among 81 crops with at least five years of history, the median peak-to-trough seasonal swing is about 29%. Post-monsoon (Oct–Nov) is the peak season for 41 of them, and Summer and Winter are the cheapest for most.
- **Headline seasonal YoY needs care.** When all reporting crops are averaged, Winter 2025 shows +56.3% YoY. Holding the crop list constant (79 crops present in both years) it is +11.6%. Crop coverage grew from about 81 crops in 2015 to 175 in 2026, so part of the headline is a mix effect, not price growth.
- **Some rankings rest on thin data.** 25 of 175 crops have fewer than 12 months of history, and 6 have a single record. Mentha oil, the highest-priced "crop" (₹1,37,347 per quintal), has one monthly observation.

> **What we recommend**
> Prioritise price-protection effort on high-volatility, high-spend crops; time purchases around the post-monsoon peak for strongly seasonal crops; and read YoY and ranking cards only alongside their data-sufficiency caveats. Section 7 gives the full recommendations with evidence and limits.

1. Business context and objectives

Agricultural commodity prices in India are noisy: they respond to weather, arrivals, storage and festivals. A buyer who looks only at the latest price misses two things that matter for planning: how far prices typically swing, and when in the year they tend to be high or low. Mandi Price Watch turns the public Agmarknet price series into an interactive Power BI dashboard so that these patterns can be explored by crop and season.

Objectives

- Describe price levels across crops in a comparable unit (₹ per quintal and ₹ per kg).
- Quantify price risk with a simple, explainable measure (coefficient of variation).
- Show seasonal behaviour and year-over-year movement by season.
- Convert the patterns into procurement-ready recommendations and be explicit about what the data cannot prove.

Dashboard structure

Page	Purpose
1. Mandi Price Analysis	KPI cards (Total Crops 175, Total Category 17, Median Market Price ₹2,391, Highest Recorded Price ₹1,62,069, Price Volatility %); donut of crops by category; Top 10 most expensive and most affordable crops; selected-crop price trend; crop-wise price comparison; slicers for Category/crop, Year and Season Category.
2. Seasonality and Volatility	Seasonal selected vs previous year price by category; Price Volatility % by crop; Seasonal YoY % by season; annual seasonal price heatmap (matrix); volatility treemap by crop and season; seasonal price comparison; slicers for Category/crop, Season Year and Season Category.
3. Executive Insights	Cards for Highest Risk Crop, Market Leader Crop, Seasonal YoY % and Best Performing Season; Crop Risk & Price Positioning scatter; Seasonal Price Performance by category; three dynamic Market Highlights sentences; three recommendation cards.

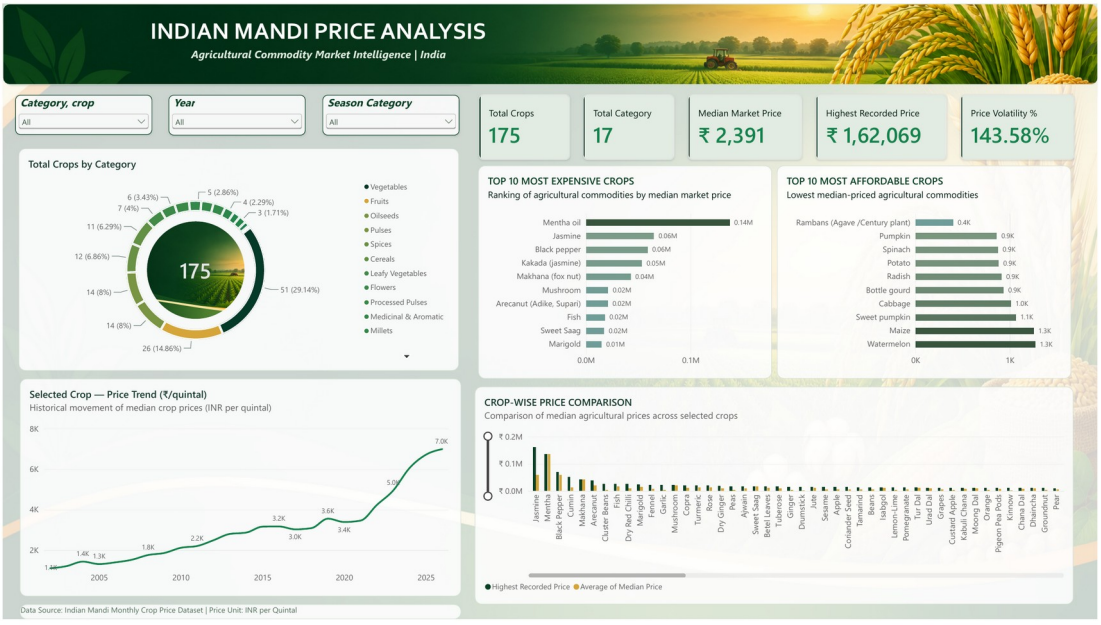


Figure A. Page 1 – Mandi Price Analysis

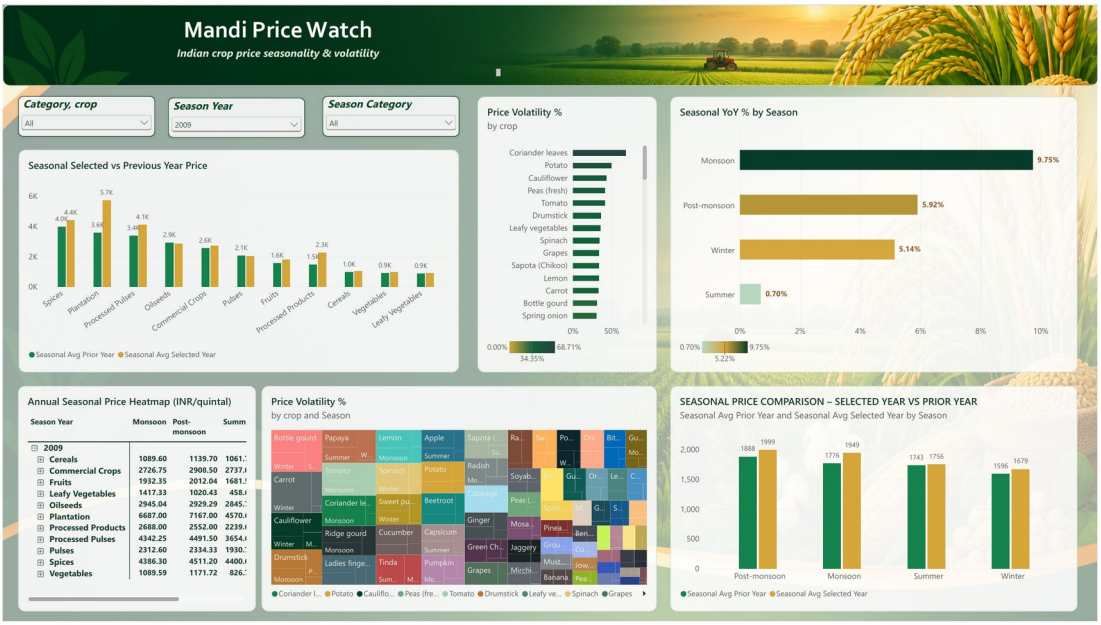


Figure B. Page 2 – Seasonality and Volatility (Season Year 2009 selected)

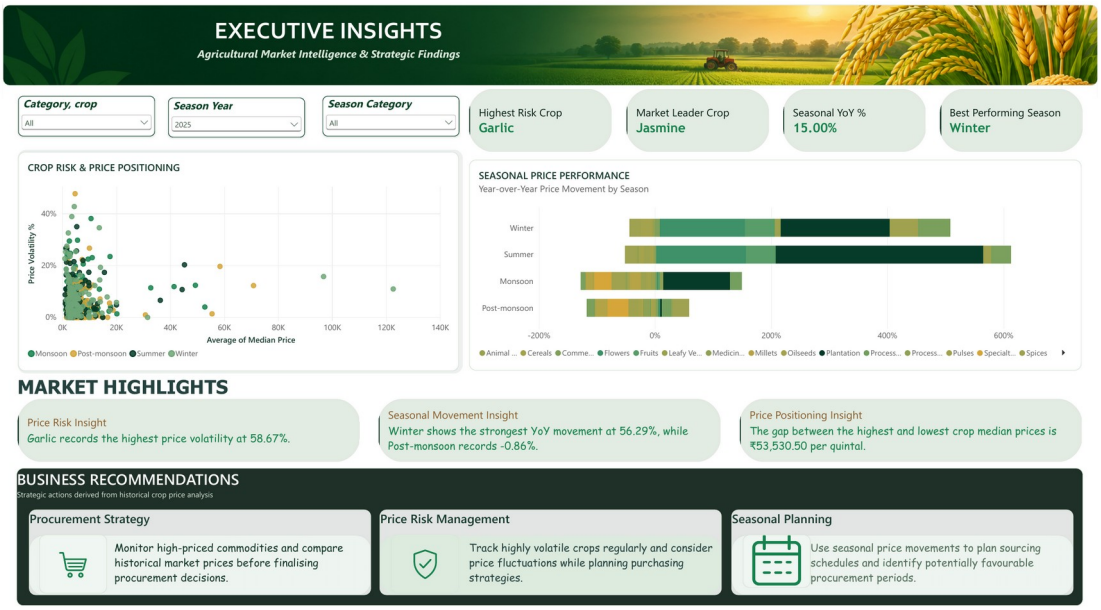


Figure C. Page 3 – Executive Insights (Season Year 2025 selected)

Reading the dashboard correctly

- **Slicer state changes the numbers.** Page 2 opens with Season Year 2009 and Page 3 with Season Year 2025. With 2025 selected, the cards show Garlic (58.67%) as Highest Risk Crop, Jasmine as Market Leader, a price gap of ₹53,530.50 per quintal, and Seasonal YoY of 15.00%. With no year selected the same measures give Carrot (82.11%), Mentha oil (one record) and a gap of ₹1,36,947. All three views reconcile with the CSV; other figures in this report use all years unless stated.
- **The Price Volatility % card on Page 1 (143.58%) is a cross-crop spread.** With no crop selected it mixes all 175 crops into one series, so it describes how different crop prices are from each other, not how much prices move over time. Read volatility per crop.
- **The Top 10 charts rank by average, not median.** Although the subtitles say “median market price”, the bars match the average of monthly median prices (for example Potato shows 0.9K,

which is its ₹883 average, against a ₹712 median). The expensive list is also led by one-record products (Mentha oil, Makhana). Relabel the charts or switch them to Median Market Price with a minimum-history rule.

2. Data and methodology

Dataset

The file contains 19,680 crop-month records for 175 crops from February 2002 to November 2026, with no blank values and no duplicate crop-month pairs. The median record count per crop is 44 months. The price is the median across markets of each market's monthly average modal price, so it is a market-level summary and not a single trade price. The Category field (17 categories, for example Vegetables with 51 crops) was added inside the Power BI model and is not part of the source CSV, so category-level results are shown only as they appear in the dashboard.

How the key measures are defined

These definitions are taken directly from the Power BI model, so the report and dashboard use identical logic.

Measure	DAX logic	Plain-English meaning
Price Volatility %	DIVIDE(STDEV.P(Median Price), Average Price)	Standard deviation as a share of the average price (coefficient of variation) over the selected crops, months and filters.
Median Market Price	MEDIAN(Median Price)	Middle observed monthly price in the selection (₹ per quintal).
Season Category	Dec–Feb Winter; Mar–May Summer; Jun–Sep Monsoon; Oct–Nov Post-monsoon	Four seasons; note that Monsoon spans four months and Post-monsoon two.
Season Year	Calendar year, +1 for December	Keeps one winter (Dec–Feb) in a single season-year.
Seasonal YoY %	(Selected-year average − prior-year average) ÷ prior-year average	Year-over-year change in the average price for the chosen season; shown only when one Season Year is selected.

Analysis approach

- Rankings use crops with at least 12 monthly records (150 of 175) so that one or two months cannot dominate a result.
- Seasonal patterns use crops with at least 60 months of data (81 crops). For each crop, prices are compared with that crop's own calendar-year average (in logs) before averaging by season, which removes most of the long-run trend.
- Correlations are Spearman rank correlations, which are robust to the very wide price range.
- YoY comparisons are shown two ways: using all crops reporting (as the dashboard does) and using only crops present in both years.

3. Data quality and coverage

Coverage is the single most important context for reading the dashboard. Only 21 crops reported in 2002, about 80 through 2023, then 114 in 2024, 154 in 2025 and 175 in 2026.

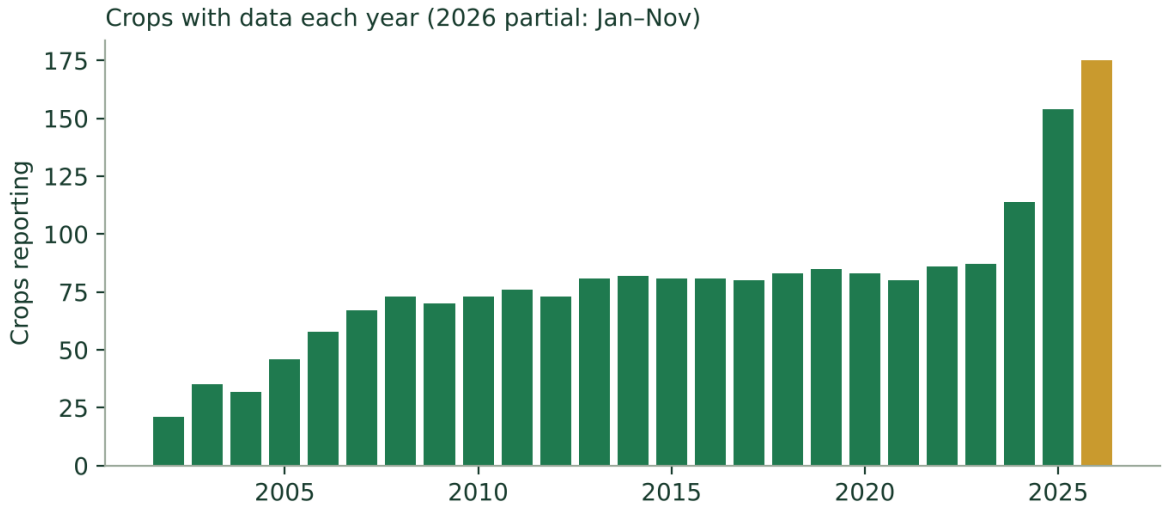


Figure 1. Number of crops with data in each year

- **Partial latest month.** November 2026 has 161 crops against 175 in October; the file was built on 3 November 2026, so Post-monsoon 2026 is incomplete.
- **Thin history.** 25 crops have fewer than 12 months, 11 have three or fewer, and 6 have a single record. These crops have a volatility of 0% or an unstable value, and should not be ranked against long-history crops.
- **Extreme values are real but rare.** The three highest records are Jasmine (₹1,62,069 in Jan 2026), Mentha oil (₹1,37,347 in Oct 2026) and Jasmine (₹1,36,452 in Jan 2025). The dataset mixes vegetables, grains, spices, flowers, oils and fish, so raw “highest price” statistics mostly compare product categories.

4. Price levels

The overall median price is ₹2,391 per quintal (₹23.91 per kg); the middle half of all records lie between ₹1,317 and ₹4,256. The mean (₹3,677) is well above the median because of a long right tail of high-value products.

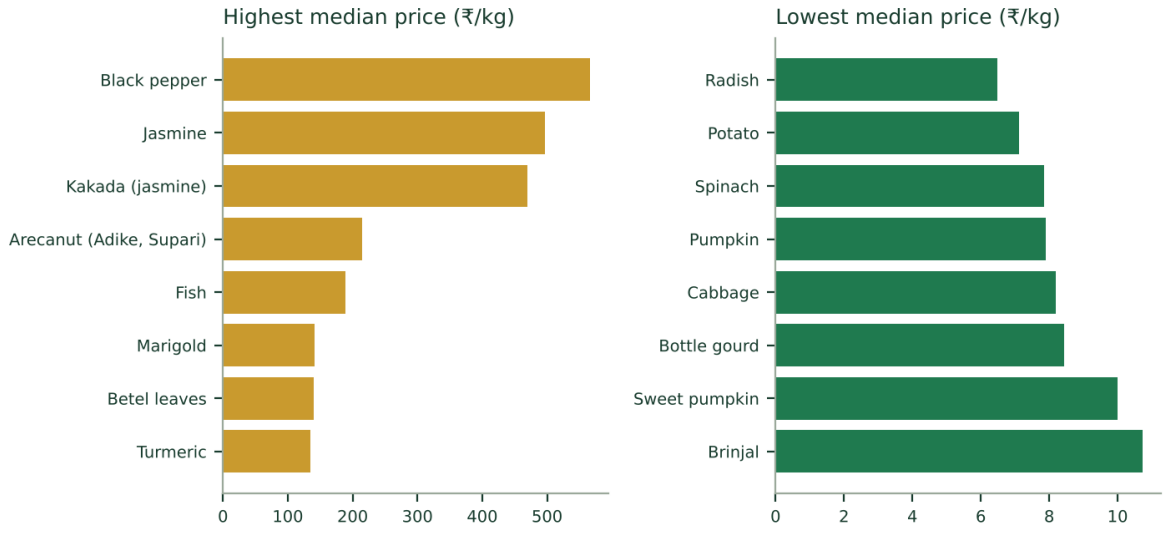


Figure 2. Highest and lowest median prices (crops with ≥12 months of data)

Among crops with at least 12 months of data, Black pepper (₹56,625 per quintal), Jasmine (₹49,706) and Kakada jasmine (₹46,955) have the highest median prices; Radish (₹649), Potato (₹712) and Spinach (₹786) are the lowest. The dashboard's Price Positioning card reports the gap between the highest and lowest crop median. Without filters that gap is ₹1,36,947 per quintal, driven by Mentha oil (one record) at one end and Rambans (₹400) at the other, so it describes product mix more than market conditions. A minimum-history rule is recommended for that card.

5. Price volatility and risk

Volatility is measured as the coefficient of variation: the standard deviation of monthly prices divided by their average. It is unit-free, so a ₹600 vegetable and a ₹50,000 spice can be compared.

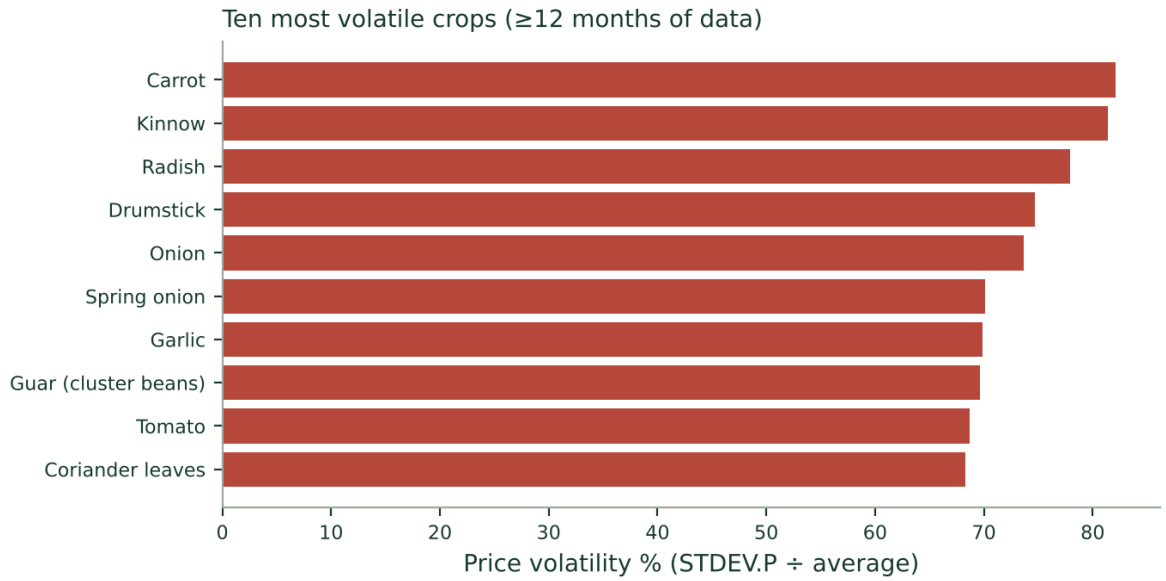


Figure 3. Most volatile crops by coefficient of variation

Crop	Volatility %	Months of data	Median ₹/quintal
Carrot	82.1%	246	1,332
Kinnow	81.4%	121	1,667
Radish	77.9%	254	649
Drumstick	74.7%	240	2,726
Onion	73.7%	290	1,171
Spring onion	70.1%	151	1,850
Garlic	69.9%	243	3,641
Guar (cluster beans)	69.6%	264	3,537
Tomato	68.7%	292	1,166
Coriander leaves	68.3%	223	1,985

Price level versus volatility

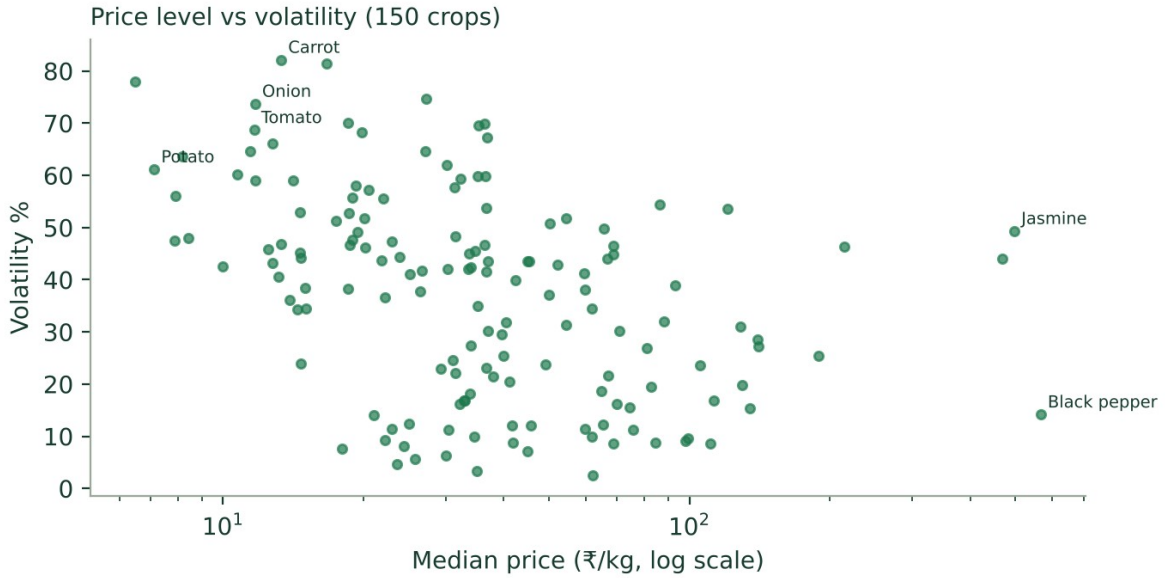


Figure 4. Median price (log scale) against volatility for 150 crops

Lower-priced crops are more volatile on average: the median volatility is 47.8% in the lowest price third of crops, 36.4% in the middle third and 27.1% in the highest. The rank correlation is −0.44 ($p < 0.001$). A plausible explanation is that cheap, perishable produce has short shelf life and supply shocks pass quickly into price, whereas higher-value dry or specialty products are stored and traded more steadily. The data supports the pattern, not the explanation.

Volatility versus trend

Because the measure uses the whole 2002–2026 period, it includes long-run price growth as well as month-to-month swings. For the 81 crops with at least five years of data, the median underlying price growth is about 6.8% per year, and volatility is positively correlated with the size of that trend (rank correlation 0.55). Removing the trend changes the ranking: Coriander leaves, Peas (fresh), Drumstick, Methi (leaves) and Green peas show the largest swings around their own trend, while Carrot and Kinnow, which top the raw ranking, move down. Both views are useful: the raw measure shows total price dispersion a buyer faced, the detrended one isolates short-term unpredictability.

6. Seasonality

Seasonal pattern by crop

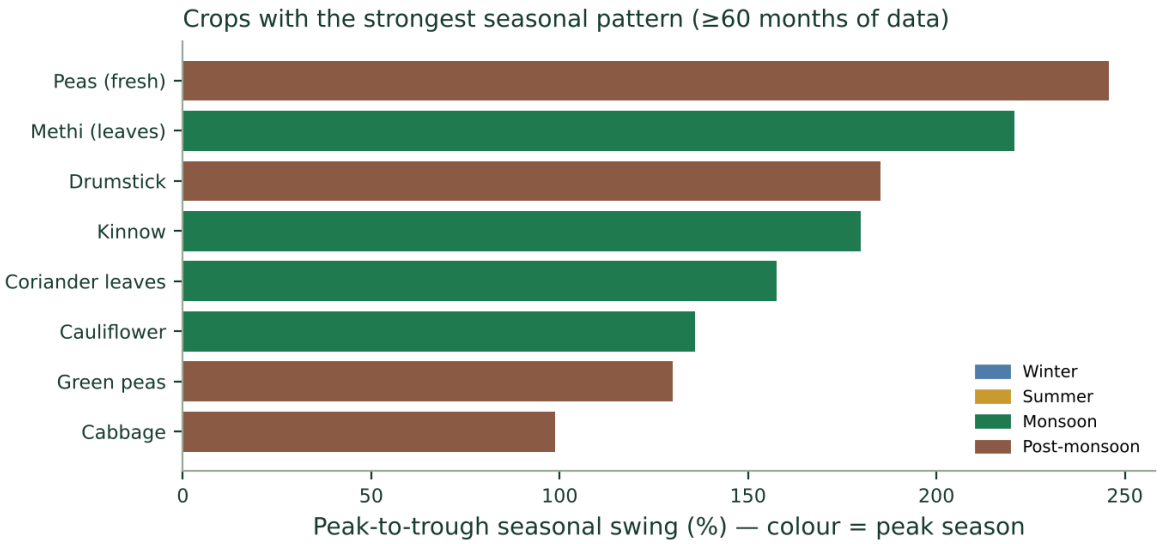


Figure 5. Crops with the strongest seasonal patterns

Using each crop's own calendar-year average as the baseline, the median peak-to-trough seasonal gap across 81 crops is about 29%. The strongest patterns are Peas (fresh) (peak Post-monsoon, trough Winter; ~246% gap), Methi leaves (Monsoon vs Winter; ~221%), Drumstick (Post-monsoon vs Summer; ~185%), Kinnow (Monsoon vs Post-monsoon; ~180%) and Coriander leaves (Monsoon vs Winter; ~157%). Winter-grown leafy and pod vegetables are cheap in their own harvest season and expensive in the off-season.

Season	Crops peaking here	Crops at their low here	Months in data
Winter (Dec–Feb)	11	30	73
Summer (Mar–May)	12	32	75
Monsoon (Jun–Sep)	17	10	100
Post-monsoon (Oct–Nov)	41	9	50

Post-monsoon is the peak season for about half of the crops studied (41 of 81), while Summer and Winter are the most common lows. Because Post-monsoon is only two months long and includes the partial November 2026, it is the least reliable of the four.

Seasonal year-over-year

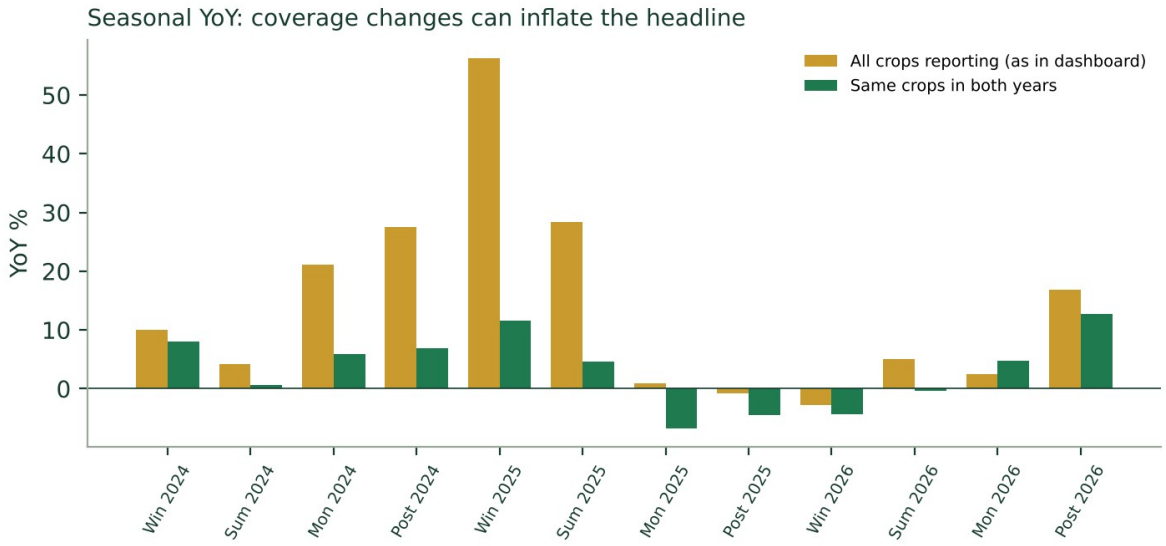


Figure 6. Seasonal YoY: all crops reporting versus the same crops in both years

Seasonal YoY depends strongly on which crops are included. In Winter 2025 the all-crops average rose 56.3% while the like-for-like figure was 11.6%; in Monsoon 2024 the gap was 21.1% against 5.9%. By 2026 the two views converge because coverage stopped expanding. On a like-for-like basis, price changes have been moderate: roughly +0.6% to +12% in 2024 and 2025 for most seasons, turning slightly negative in Monsoon and Post-monsoon 2025 and in Winter 2026, then recovering in Monsoon and Post-monsoon 2026 (the latter still incomplete).

7. Business recommendations

The three dashboard cards map to the findings above. Each recommendation states its evidence and what to verify before acting.

A. Procurement strategy

- **Action:** Segment crops by volatility and spend. For high-volatility crops with large purchase volumes (for example onion, tomato, garlic), use phased buying and price-locking arrangements; for steadier crops, regular spot buying is sufficient.
- **Evidence:** Volatility is highest in carrot, radish, onion, tomato and similar produce, and falls as price level rises.
- **Check first:** Volatility % ignores how much is bought. A very volatile crop with a tiny budget may matter less than a moderately volatile crop with a large one, so combine this ranking with purchase volumes.

B. Price risk management

- **Action:** Track volatility for each crop over a fixed recent window, with a minimum of 12 months of history, and set review triggers when the latest price moves outside its normal range. Read YoY on a like-for-like crop set.
- **Evidence:** The all-crops YoY overstated Winter 2025 growth by about 45 percentage points, and 25 crops have too little history to rank.
- **Check first:** Whether detrended volatility changes which crops you consider riskiest (section 5).

C. Seasonal planning

- **Action:** For crops with a strong, repeatable seasonal pattern (such as fresh peas, methi, coriander leaves, cauliflower), plan purchases and contracts around the low season and avoid last-minute buying in the peak season. Evaluate storage or processed alternatives where the product allows it.
- **Evidence:** Median seasonal swing of about 29% across 81 crops, up to more than 100% for the leading seasonal crops; Post-monsoon is the peak for about half of them.
- **Check first:** Regional timing differs across India. The national median across markets can blur local harvest windows, so validate against the specific mandis you buy from.

8. What the data supports and what it does not

Conclusion	Status
Cheap perishables (carrot, radish, onion, tomato) are the most volatile crops	Supported by the data (150 crops, ≥12 months each).
Higher-priced crops are less volatile on average	Supported as an association (ρ = −0.44); the cause is a hypothesis.
Strong seasonal patterns exist; Post-monsoon is the typical peak	Supported for 81 long-history crops; Post-monsoon is short and the latest month is partial.
All-crops seasonal YoY reflects price growth	Not supported: coverage growth inflates it; use a like-for-like comparison.
Highest-priced “crop” and the price gap describe market conditions	Not supported: they are driven by one-record products and by mixing categories.
Volatility measures forward-looking risk	Needs investigation: the measure includes the long-run trend and is historical.
Recent price direction by crop	Needs investigation: coverage and month availability differ between recent windows.

9. Limitations and next steps

- Prices are national medians of market averages; they hide regional differences and quality grades, and they are not transaction prices.
- Coverage changes over time and many crops have short histories, so crop-to-crop comparisons are not on equal footing.
- The coefficient of variation uses the whole selected period and is sensitive to trend and to extreme observations; detrended or rolling-window versions would be more forward-looking.
- Season definitions are uneven in length (Monsoon four months, Post-monsoon two) and November 2026 is a partial month.
- Next steps: add a minimum-record rule to the ranking cards, add a like-for-like YoY measure, add rolling 12-month volatility, and join arrival volumes or weather data to test the seasonal and volatility explanations.

Reproducibility: all statistics in this report were computed with Python (pandas, SciPy) from the supplied CSV, using the definitions in section 2. Figures follow the Power BI measures, except where stated (detrended volatility, like-for-like YoY and seasonal indices are additional analyses).